TUTORIAL 12

Similarity, Eigenvalues, and the Characteristic Polynomial

MOTIVATION AND GUIDING IDEA

A change of basis preserves the linear map but can greatly simplify its matrix. The determinant tests invertibility without computing an inverse, detects dependence among rows, and measures changes in area or volume. Before a long calculation, look for zeros, linear dependence, or block structure.

NOTATION AND CONVENTIONS

- $M_n(\mathbb{F})$ is the space of square matrices of size n over $\mathbb{F}$; I_n is the identity matrix, and A^T is the transpose.
- $A \sim B$ means that $B = P^{-1}AP$ for some invertible P. The columns of P are the new basis vectors expressed in the old basis.
- $\det A$ is the determinant. The alternative notation $|A|$ sometimes means the same scalar; it is not an absolute value of a matrix.
- M_{ij} is obtained by deleting row i and column j; $C_{ij} = (-1)^{i+j} \det(M_{ij})$ is the corresponding cofactor.
- $R_i \leftarrow R_i + cR_j$ means replacing row i by itself plus c times row j. The determinant preservation rule requires $i \neq j$.

NOTE

Distinguish two transformations. Similarity, $B = P^{-1}AP$, describes an operator after the same basis change in domain and codomain; it preserves the determinant. Congruence, $B = P^TAP$, describes a basis change for a bilinear form; here $\det B = (\det P)^2 \det A$. The formulas cannot be interchanged.

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12.1 Similar matrices

RECALL AND TOOLS

Matrices $A, B \in \mathbb{F}^{n\times n}$ are similar if there is an invertible $P \in \mathbb{F}^{n\times n}$ such that

$$A = P^{-1}BP$$

If A, B are similar, then

1. $$\det(A) = \det(B)$$
2. $$\text{trace}(A) = \text{trace}(B)$$
3. $$\text{rank}(A) = \text{rank}(B)$$

EXERCISE 12.1

$$A, B \in \mathbb{F}^{n\times n}.$$

Please prove or disprove: if A, B are similar, then A^k, B^k are similar for every $k \geq 1$; if A^k, B^k are similar for some $k \geq 1$, then A, B are similar.

WORKED SOLUTION

The first claim is true: cancelling adjacent inverse factors gives similarity through the same P. The second claim is false over every field. The displayed cubes agree, but a matrix similar to zero must be zero, whereas B is nonzero.

$$B = P^{-1}AP \quad \Longrightarrow \quad B^k = (P^{-1}AP)^k = P^{-1}A^kP.$$

$$A = 0_{3\times 3}, \qquad B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

$$A^3 = B^3 = 0, \qquad B^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad B \neq 0.$$

EXERCISE 12.2

$$T : \mathbb{R}^2 \to \mathbb{R}^2, \quad T(x, y)^T = (2x + y, x - y)^T, \qquad B = ((1, 1)^T, (1, -2)^T).$$

Please compute $[T]_E^B$, please find v with $T(v) = (9, -6)^T$, and please check the result in both coordinate systems.

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EXERCISE 12.2 CONTINUED

WORKED SOLUTION

The columns are the images in standard output coordinates. Dividing by 3 gives the input B-coordinates, and rebuilding in B gives the vector itself.

$$T(1,1)^T = (3,0)^T, \qquad T(1,-2)^T = (0,3)^T, \qquad [T]_E^B = 3I_2.$$

$$[v]_B = (3,-2)^T, \qquad v = 3(1,1)^T - 2(1,-2)^T = (1,7)^T.$$

$$T(1,7)^T = (2+7, 1-7)^T = (9,-6)^T, \qquad 3I_2(3,-2)^T = (9,-6)^T.$$

12.2 Similar Matrices

RECALL AND TOOLS

Matrices $A, B \in \mathbb{F}^{n\times n}$ are called similar if there is an invertible matrix $P \in \mathbb{F}^{n\times n}$ such that

$$A = P^{-1}BP$$

If A, B are similar, then

1. $$\det(A) = \det(B)$$
2. $$\operatorname{trace}(A) = \operatorname{trace}(B)$$
3. $$\operatorname{rank}(A) = \operatorname{rank}(B)$$

EXERCISE 12.3

Let $A, B \in \mathbb{F}^{n\times n}$. Please prove or disprove:

1. If A, B are similar, then A^k, B^k are similar for every integer $k \geq 1$.
2. If A^k, B^k are similar for some integer $k \geq 1$, then A, B are similar.

WORKED SOLUTION

1. If $B = P^{-1}AP$ with P invertible, cancellation using $PP^{-1} = I$ gives

   $$B^k = (P^{-1}AP)^k = P^{-1}A^kP.$$

   Thus the powers are similar via the same matrix P.

2. The statement is false. Over any field, take

   $$A = 0_{2\times 2}, \qquad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

   We have $A^2 = B^2 = 0$, but A, B are not similar: $P^{-1}AP = 0$ for every invertible P,

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EXERCISE 12.3 CONTINUED

whereas $B \neq 0$. Over $\mathbb{R}$ or $\mathbb{C}$, the example $A = -2I_2$, $B = 2I_2$ also works: their squares are equal, and a scalar matrix is similar only to itself. This example gives no counterexample in characteristic 2, where $-2I_2 = 2I_2$.

Common pitfall: The equality $A^k = (Q^{-1}BQ)^k$ does not imply $A = Q^{-1}BQ$. Taking a power is not an injective operation on matrices, as these examples show.

12.3 Similar Matrices

DEFINITION 12.1

Let $A, B \in M_n(\mathbb{F})$.
We say that A is **similar** to B (and write $A \sim B$) if there is an invertible $P \in M_n(\mathbb{F})$ such that

$$A = P^{-1}BP$$

PROPOSITION 12.1

Let $A, B \in M_n(\mathbb{F})$. They are similar if and only if there exist an n-dimensional vector space V, a linear operator $T : V \to V$, and two ordered bases $\mathcal{E}, \mathcal{F}$ such that

$$A = [T]_{\mathcal{E}}, \qquad B = [T]_{\mathcal{F}}.$$

If P converts coordinates from $\mathcal{E}$ to $\mathcal{F}$, so that $[v]_{\mathcal{F}} = P[v]_{\mathcal{E}}$, then

$$A = P^{-1}BP.$$

Thus similarity describes the same operation in two coordinate systems.

EXERCISE 12.4 EXAM QUESTION

Please prove or disprove the following statements.

1. There do not exist fields $\mathbb{F}_1 \subseteq \mathbb{F}_2$ and a vector space V over $\mathbb{F}_2$, also regarded as an $\mathbb{F}_1$-vector space by restricting scalars, such that

$$\dim_{\mathbb{F}_1} V = 2 \dim_{\mathbb{F}_2} V.$$

2. If $A \in M_n(\mathbb{F})$ is invertible, then every $B \in M_{n\times m}(\mathbb{F})$ satisfies

$$N(AB) = N(B).$$

3. Let $n \geq 1$. For every integer k with $0 \leq k \leq n^2$, there exists $A \in M_n(\mathbb{R})$ with $\mathrm{tr}(A) = k$ that is similar only to itself.

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EXERCISE 12.4 EXAM QUESTION CONTINUED

WORKED SOLUTION

1. The claim is false. Take $\mathbb{F}_1 = \mathbb{R}$, $\mathbb{F}_2 = \mathbb{C}$ and $V = \mathbb{C}^2$. Appropriate bases are

$$B_{\mathbb{C}} = ((1,0),(0,1)), \qquad B_{\mathbb{R}} = ((1,0),(i,0),(0,1),(0,i)).$$

$$\dim_{\mathbb{C}} V = 2, \qquad \dim_{\mathbb{R}} V = 4 = 2\dim_{\mathbb{C}} V.$$

2. True. Prove both inclusions. If $x \in N(B)$, then $Bx = 0$, and

$$ABx = A(Bx) = A0 = 0.$$

so $x \in N(AB)$. Hence $N(B) \subseteq N(AB)$. Conversely, if $x \in N(AB)$, then $ABx = 0$. Multiplication on the left by A^{-1} gives

$$Bx = (A^{-1}A)Bx = A^{-1}0 = 0.$$

so $x \in N(B)$, proving $N(AB) \subseteq N(B)$. Together the inclusions give the required equality.

3. True. Define

$$A = \frac{k}{n} I_n.$$

Since $n \geq 1$ and the field is $\mathbb{R}$, the quotient is defined, and

$$\operatorname{tr}(A) = \sum_{j=1}^{n} \frac{k}{n} = n\frac{k}{n} = k.$$

For every invertible $P \in M_n(\mathbb{R})$,

$$P^{-1}AP = P^{-1}\left(\frac{k}{n} I_n\right) P = \frac{k}{n} P^{-1}P = \frac{k}{n} I_n = A.$$

Thus every matrix similar to A equals A. In fact this construction works for every real k, including all the values in the statement.

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12.4 Eigenvalues and Eigenvectors

12.4.1 Motivation and Background

MOTIVATION AND GUIDING IDEA

Eigenvalues and eigenvectors help simplify complex systems. Linear operators (and matrices) generally rotate vectors and mix their coordinates. We seek nonzero vectors on which the action is multiplication by a scalar. Such vectors are **eigenvectors** , and the scalar is an **eigenvalue** . Over $\mathbb{R}$, a positive eigenvalue stretches or contracts along the same direction, a negative one also reverses direction, and a zero eigenvalue sends the vector to zero.

Eigenvalues describe models throughout engineering. The following examples show how one algebraic equation appears in different settings:

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- **Google's ranking algorithm (PageRank):** In a random browsing model, a transition matrix M sends a probability distribution to the next one. A stationary ranking distribution $\vec{r}$ satisfies

$$M\vec{r} = \vec{r}, \qquad r_i \geq 0, \qquad \sum_i r_i = 1.$$

  This is an eigenvalue equation with $\lambda = 1$. The full PageRank model adds a random jump mechanism to address the link structure; the exercise below studies a simplified model.

- **Quantum mechanics:** In a finite-dimensional quantum model, a pure state is a normalized vector and a Hermitian Hamiltonian H represents energy. The eigenvalue equation

$$H\vec{\psi} = E\vec{\psi}$$

  identifies states of definite energy and possible energy values. It is an application of the same linear equation; a general description of mixed states requires the additional concept of a density matrix.

- **Systems of differential equations:** If A is diagonalizable and $A = PDP^{-1}$, substituting $\vec{y} = P\vec{z}$ into $\vec{y}' = A\vec{y}$ gives

$$P\vec{z}' = AP\vec{z} = PD\vec{z} \quad \Longrightarrow \quad \vec{z}' = D\vec{z}.$$

  Each component satisfies $z_i' = \lambda_i z_i$, so $z_i(t) = z_i(0)e^{\lambda_i t}$. For a continuous linear system, all solutions decay to zero if every eigenvalue has negative real part. If an eigenvalue has positive real part, some solutions grow. A zero real part requires further analysis; its sign alone does not imply decay.

- **Classifying critical points in calculus:** For a twice continuously differentiable real function, consider the symmetric Hessian at a critical point. If all its eigenvalues are positive, the point is a strict local minimum; if all are negative, it is a strict local maximum. Positive and negative eigenvalues imply a saddle point. When zero eigenvalues occur without both signs, the second derivative test may be inconclusive.

DEFINITION 12.2

Let V be a vector space over a field $\mathbb{F}$, and let $T : V \to V$ be a linear map (a linear operator). A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of T if there is a **nonzero** vector $\vec{v} \in V$ (that is, $\vec{v} \neq \vec{0}_V$) such that

$$T(\vec{v}) = \lambda\vec{v}$$

This is the **eigenvalue equation** . The vector $\vec{v}$ is an **eigenvector** of T associated with λ.

DEFINITION 12.3

Let $A \in \mathbb{F}^{n\times n}$ be a square matrix of size n.
A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of A if there is a **nonzero** column vector $\vec{x} \in \mathbb{F}^n$ (that is, $\vec{x} \neq \vec{0}$) such that

$$A\vec{x} = \lambda\vec{x}$$

This is the **eigenvalue equation** . The vector $\vec{x}$ is an **eigenvector** of A associated with λ.

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NOTE

Over $\mathbb{R}$, the line span$\{\vec{x}\}$ is invariant. If $\lambda < 0$, direction is reversed; if $\lambda = 0$, the image is the zero vector. The algebraic concept also applies over $\mathbb{C}$, but a complex eigenvalue cannot always be interpreted as stretching a real line.

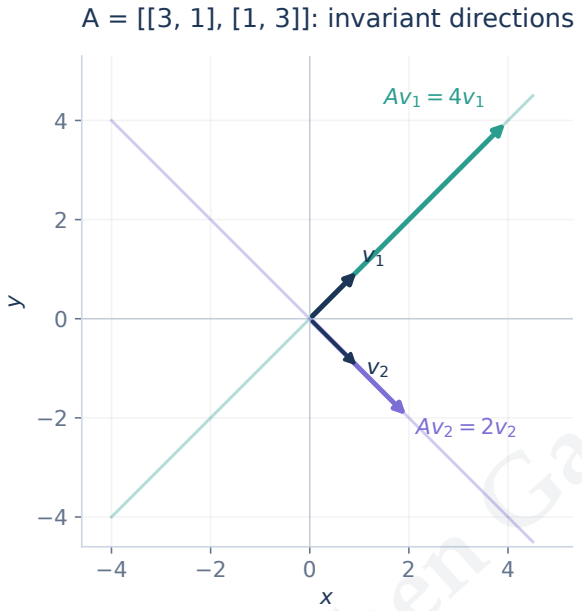


Figure 12.1. Vectors in eigendirections remain on the lines they span. A general vector may change direction because its components are stretched by different factors.

12.5 The Characteristic Polynomial

DEFINITION **12.4**

For $A \in M_n(F)$, the characteristic polynomial is $p_A(t) = \det(tI_n - A)$, where t is a polynomial variable and I_n the identity matrix. There is an eigenvector for $\lambda \in F$ if and only if $\lambda I_n - A$ is singular, equivalently $p_A(\lambda) = 0$. The solution space of $Av = \lambda v$ is the eigenspace $E_\lambda = \ker(A - \lambda I_n)$. It contains zero, whereas an eigenvector is required to be nonzero.

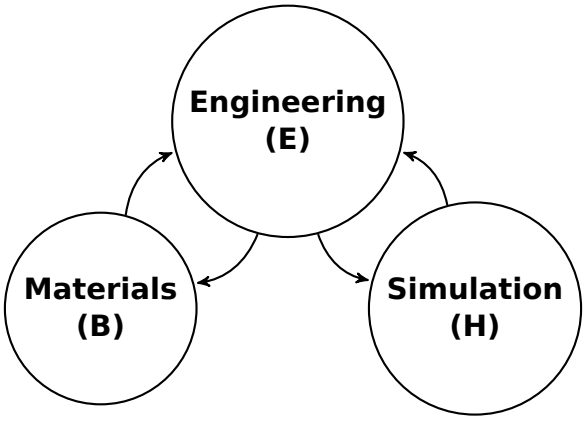
EXERCISE **12.5**

Three sites B, E, H are linked as follows. From either B or H, the next site is always E. From E, the next site is B with probability $1/3$ and H with probability $2/3$.

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EXERCISE 12.5 CONTINUED



Please find a stationary ranking distribution whose entries sum to 1, and please compare the three rankings.

WORKED SOLUTION

In the order (B, E, H), the transition matrix is

$$M = \begin{pmatrix} 0 & 1/3 & 0 \\ 1 & 0 & 1 \\ 0 & 2/3 & 0 \end{pmatrix}.$$

The equation $Mr = r$ gives

$$r_B = \tfrac{1}{3} r_E, \qquad r_H = \tfrac{2}{3} r_E, \qquad r_B + r_H = r_E.$$

Thus $1 = r_B + r_E + r_H = 2r_E$, so

$$r = (1/6, 1/2, 1/3)^T.$$

Site E has three times the rank of B and $3/2$ times the rank of H. A stationary distribution does not guarantee convergence from every start. Here

$$(0,1,0)^T \longmapsto (1/3, 0, 2/3)^T \longmapsto (0,1,0)^T,$$

so this initial distribution produces a cycle of length 2.

EXERCISE 12.6

On the space V of all real sequences, define

$$T(x_1, x_2, x_3, \ldots) = (x_1, 3x_2, 5x_3, \ldots), \qquad (Tx)_n = (2n-1)x_n.$$

Please find all eigenvalues and all eigenvectors of T.

WORKED SOLUTION

In each coordinate, the eigenvalue equation is

$$(2n-1)x_n = \lambda x_n, \qquad (2n - 1 - \lambda)x_n = 0.$$

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EXERCISE 12.6 CONTINUED

If λ is not a positive odd integer, every coefficient is nonzero, so $x = 0$. For $\lambda = 2m - 1$, only coordinate m can be nonzero. Therefore

$$\lambda_m = 2m - 1, \qquad E_{\lambda_m} = \text{span}\{e_m\}, \qquad m = 1, 2, \ldots.$$

All eigenvectors are ce_m with $c \neq 0$. For example, $Te_1 = e_1$ and $Te_2 = 3e_2$.

NOTE

The exercise concerns all sequences, with no summability or norm assumption. The unit sequences $\{e_n\}$ are not an algebraic basis of this space: finite linear combinations give only finitely supported sequences, not, for example, $(1, 1, 1, \ldots)$. An algebraically similar model can use indices $n = 0, 1, 2, \ldots$ and the rule $Ne_n = ne_n$. This resembles a number operator. The exercise alone does not imply that all physical energies are discrete or that every physical operator acts on the space of all sequences.

EXERCISE 12.7

Let $n \geq 2$, let $V = M_n(\mathbb{C})$ be the space of square matrices, and let $p, q \in \mathbb{C}$ with $q \neq 0$. Define the linear operator $T : V \to V$ by

$$T(A) = pA + qA^T$$

Please find the eigenvalues and eigenvectors of T directly from the eigenvalue equation.

WORKED SOLUTION

Let $A \neq 0$ be an eigenvector of T with eigenvalue λ. Then

$$pA + qA^T = \lambda A \tag{12.1}$$

Isolate the transpose term:

$$qA^T = (\lambda - p)A$$

Divide by q (which is nonzero) to express A^T in terms of A:

$$A^T = \frac{\lambda - p}{q} A \tag{12.2}$$

Transpose both sides of (12.2):

$$(A^T)^T = \left(\frac{\lambda - p}{q} A\right)^T$$

Using the properties of transposition,

$$A = \frac{\lambda - p}{q} A^T$$

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EXERCISE 12.7 CONTINUED

Substitute the expression for A^T from (12.2) into the last equation:

$$A = \frac{\lambda - p}{q} \cdot \underbrace{\left(\frac{\lambda - p}{q} A\right)}_{A^T}$$

$$A = \left(\frac{\lambda - p}{q}\right)^2 A$$

Rearrange and factor out A on the right:

$$\left[1 - \left(\frac{\lambda - p}{q}\right)^2\right] A = 0$$

Since $A \neq 0$ (an eigenvector), the scalar in brackets must vanish, so $\left(\frac{\lambda-p}{q}\right)^2 = 1$. There are two cases:

1. **First case:** $\frac{\lambda-p}{q} = 1$. Solving for λ gives the first eigenvalue:

   $$\lambda_1 = p + q$$

   To identify its eigenvectors, substitute $\frac{\lambda-p}{q} = 1$ directly into (12.2):

   $$A^T = 1 \cdot A \implies A^T = A$$

   Thus the eigenvectors for λ_1 are precisely the nonzero symmetric matrices.

2. **Second case:** $\frac{\lambda-p}{q} = -1$. Solving for λ gives the second eigenvalue:

   $$\lambda_2 = p - q$$

   Substitute $\frac{\lambda-p}{q} = -1$ into (12.2):

   $$A^T = -1 \cdot A \implies A^T = -A$$

   Thus the eigenvectors for λ_2 are precisely the nonzero skew-symmetric matrices.

DEFINITION 12.5

Definition: the eigenspace V_λ

Let λ be an eigenvalue. The eigenspace V_λ is the set of all eigenvectors for λ together with the zero vector. The zero vector belongs to this subspace but is not an eigenvector. It follows directly from the eigenvalue equation:

1. **For a linear operator T:** Starting from $T(\vec{v}) = \lambda\vec{v}$, rearrange:

   $$T(\vec{v}) - \lambda\vec{v} = \vec{0} \implies T(\vec{v}) - \lambda I_V(\vec{v}) = \vec{0}$$

   Factor:

   $$(T - \lambda I_V)\vec{v} = \vec{0}$$

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Thus the eigenspace is the kernel of the operator

$$V_\lambda = \ker(T - \lambda I_V)$$

2. **For a matrix A:** Starting from $A\vec{v} = \lambda\vec{v}$, rearrange:

$$\lambda\vec{v} - A\vec{v} = \vec{0} \implies \lambda I_n\vec{v} - A\vec{v} = \vec{0}$$

Factor out $\vec{v}$ on the right:

$$(\lambda I_n - A)\vec{v} = \vec{0}$$

Thus the eigenspace is the nullspace of the matrix:

$$V_\lambda = N(\lambda I_n - A)$$

12.6 An Introduction to Diagonalization

DEFINITION 12.6

A matrix $A \in M_n(\mathbb{F})$ is diagonalizable over $\mathbb{F}$ if there are an invertible matrix $P \in M_n(\mathbb{F})$ and a diagonal matrix $D \in M_n(\mathbb{F})$ such that $A = PDP^{-1}$. If p_j is column j of P and d_j is diagonal entry j of D, then $AP = PD$ means precisely

$$Ap_j = d_j p_j \qquad (1 \le j \le n).$$

Invertibility of P means its columns form a basis. Thus diagonalization amounts to choosing a basis of eigenvectors. In those coordinates, each component is multiplied by its corresponding scalar.

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EXAMPLE 12.1

Over $\mathbb{R}$, consider

$$A = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}.$$

We have

$$A\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 10 \\ 5 \end{pmatrix} = 5\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad A\begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The vectors are independent because their column matrix has determinant -5. Hence

$$P = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix}, \quad P^{-1} = \frac{1}{5}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}.$$

The inverse can be checked using $P^2 = 5I_2$. Also,

$$AP = \begin{pmatrix} 10 & 0 \\ 5 & 0 \end{pmatrix} = PD, \qquad PDP^{-1} = \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} = A.$$

Each vector has a unique decomposition $x = \alpha(2,1)^T + \beta(1,-2)^T$, and

$$Ax = 5\alpha(2,1)^T.$$

Thus the action in the new coordinates is $(\alpha, \beta)^T \mapsto (5\alpha, 0)^T$.

EXAMPLE 12.2

Over $\mathbb{R}$, the matrix

$$J = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

has no basis of eigenvectors. Its eigenvalue equation is

$$(J - \lambda I)\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} (2-\lambda)x + y \\ (2-\lambda)y \end{pmatrix} = 0.$$

If $\lambda \neq 2$, the second equation gives $y = 0$, and the first then gives $x = 0$. Thus the only eigenvalue is 2. For that value, $y = 0$, so all eigenvectors lie on $\text{span}\{(1,0)^T\}$. Two independent vectors cannot be chosen from that line. Hence J is not diagonalizable.

EXERCISE 12.8 CHALLENGE

$$A = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}, \quad B = ((1,1)^T, (1,-1)^T), \quad x_{k+1} = Ax_k, \quad x_0 = (2,1)^T.$$

Please compute $D = [T]_B$ for $T(x) = Ax$ and $P = [I]_E^B$. Please prove $A^k = PD^kP^{-1}$ for every $k \geq 0$ and please solve the recurrence.

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EXERCISE 12.8 CHALLENGE CONTINUED

WORKED SOLUTION

The columns of P are the new basis vectors in standard coordinates. Cancelling adjacent $P^{-1}P$ factors proves the power identity; for $k = 0$ both sides are the identity. Induction on the recurrence gives $x_k = A^k x_0$. In the basis B, the two components are multiplied separately by 6 and 2.

$$A(1,1)^T = 6(1,1)^T, \quad A(1,-1)^T = 2(1,-1)^T,$$

$$D = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad A = PDP^{-1},$$

$$A^k = PD^kP^{-1} = \frac{1}{2}\begin{pmatrix} 6^k + 2^k & 6^k - 2^k \\ 6^k - 2^k & 6^k + 2^k \end{pmatrix},$$

$$x_k = A^k x_0 = \frac{1}{2}\begin{pmatrix} 3\cdot 6^k + 2^k \\ 3\cdot 6^k - 2^k \end{pmatrix}, \quad x_0 = \frac{1}{2}(4,2)^T = (2,1)^T.$$

EXERCISE 12.9

1. Let $n \geq 2$ and let $w \in \mathbb{C}$ be a primitive nth root of unity: $w^n = 1$ and $w^k \neq 1$ for every $1 \leq k \leq n-1$. Define the matrix

$$F = \left(w^{(i-1)(j-1)}\right)_{i,j=1}^{n} \in M_n(\mathbb{C}).$$

   Please compute $\det(F)$ explicitly.

2. Let $n \geq 3$, let $1 \leq p \leq n-1$, and please write $q = n - p$. For $\alpha, \beta \in \mathbb{R}$, define $A(\alpha,\beta) \in M_n(\mathbb{R})$ by

$$a_{ii} = 1 \quad (1 \leq i \leq n),$$

   and, for every $i \neq j$,

$$a_{ij} = \begin{cases} \alpha, & 1 \leq i,j \leq p \text{ or } p+1 \leq i,j \leq n, \\ \beta, & \text{otherwise.} \end{cases}$$

   Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be the linear map $T(v) = A(\alpha,\beta)v$. Please determine for which α, β the map T is an isomorphism, and please compute $\det(A(\alpha,\beta))$.

WORKED SOLUTION

First display the structure of the matrices:

$$F = \left(w^{(i-1)(j-1)}\right)_{i,j=1}^{n} = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & w & w^2 & \cdots & w^{n-1} \\ 1 & w^2 & w^4 & \cdots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^{n-1} & w^{2(n-1)} & \cdots & w^{(n-1)(n-1)} \end{pmatrix}.$$

$$A(\alpha,\beta) = \left(\begin{array}{c|c} A_{11} & A_{12} \\ \hline A_{21} & A_{22} \end{array}\right),.$$

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EXERCISE 12.9 CONTINUED

The block sizes are

$$A_{11} \in M_p(\mathbb{R}), \qquad A_{22} \in M_q(\mathbb{R}),$$
$$A_{12} \in M_{p\times q}(\mathbb{R}), \qquad A_{21} \in M_{q\times p}(\mathbb{R}).$$

$$A_{11} = \begin{pmatrix} 1 & \alpha & \alpha & \cdots & \alpha \\ \alpha & 1 & \alpha & \cdots & \alpha \\ \alpha & \alpha & 1 & \cdots & \alpha \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha & \alpha & \alpha & \cdots & 1 \end{pmatrix}_{p\times p}, \qquad A_{22} = \begin{pmatrix} 1 & \alpha & \alpha & \cdots & \alpha \\ \alpha & 1 & \alpha & \cdots & \alpha \\ \alpha & \alpha & 1 & \cdots & \alpha \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha & \alpha & \alpha & \cdots & 1 \end{pmatrix}_{q\times q}.$$

$$A_{12} = \begin{pmatrix} \beta & \beta & \cdots & \beta \\ \beta & \beta & \cdots & \beta \\ \vdots & \vdots & \ddots & \vdots \\ \beta & \beta & \cdots & \beta \end{pmatrix}_{p\times q}, \qquad A_{21} = \begin{pmatrix} \beta & \beta & \cdots & \beta \\ \beta & \beta & \cdots & \beta \\ \vdots & \vdots & \ddots & \vdots \\ \beta & \beta & \cdots & \beta \end{pmatrix}_{q\times p}.$$

$$P^{-1}A(\alpha,\beta)P = (1-\alpha)I_{n-2} \oplus B = \left(\begin{array}{c|c} (1-\alpha)I_{n-2} & 0 \\ \hline 0 & B \end{array}\right),$$

where the lower-right block has size 2:

$$B = \begin{pmatrix} 1+(p-1)\alpha & q\beta \\ p\beta & 1+(q-1)\alpha \end{pmatrix}.$$

1. **Fact used below (Vandermonde determinant).** For $x_1, \ldots, x_n \in \mathbb{C}$, define

$$V(x_1, \ldots, x_n) = \left(x_i^{j-1}\right)_{i,j=1}^{n}.$$

Then

$$\det\big(V(x_1, \ldots, x_n)\big) = \prod_{1\le i<j\le n} (x_j - x_i).$$

Set

$$x_i = w^{i-1} \qquad (i = 1, \ldots, n).$$

For every i, j,

$$F_{ij} = w^{(i-1)(j-1)} = (w^{i-1})^{j-1} = x_i^{j-1}.$$

Therefore

$$F = V(x_1, \ldots, x_n).$$

By the Vandermonde determinant formula,

$$\det(F) = \prod_{1\le i<j\le n} (x_j - x_i) = \prod_{0\le i<j\le n-1} \left(w^j - w^i\right).$$

For every $0 \le i < j \le n-1$,

$$w^j - w^i = w^i\left(w^{j-i} - 1\right),$$

so

$$\det(F) = \left(\prod_{0\le i<j\le n-1} w^i\right)\left(\prod_{0\le i<j\le n-1} (w^{j-i} - 1)\right).$$

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EXERCISE 12.9 CONTINUED

Write

$$S = \sum_{0 \le i < j \le n-1} i.$$

Then

$$\prod_{0 \le i < j \le n-1} w^i = w^S.$$

We now compute S: For each $i \in \{0, \dots, n-2\}$, there are exactly $(n-1-i)$ values of j with $i < j \le n-1$, so

$$S = \sum_{i=0}^{n-2} i(n-1-i) = \frac{n(n-1)(n-2)}{6}.$$

Consequently,

$$\prod_{0 \le i < j \le n-1} w^i = w^{\frac{n(n-1)(n-2)}{6}}.$$

Also group the factors by $m = j - i \in \{1, \dots, n-1\}$. For each m, there are exactly $(n-m)$ pairs (i, j) with $j - i = m$, so

$$\prod_{0 \le i < j \le n-1} (w^{j-i} - 1) = \prod_{m=1}^{n-1} (w^m - 1)^{n-m}.$$

Therefore

$$\boxed{\det(F) = w^{\frac{n(n-1)(n-2)}{6}} \prod_{m=1}^{n-1} (w^m - 1)^{n-m}}.$$

Because w is a primitive nth root of unity, $w^m \neq 1$ for $1 \le m \le n-1$, so $w^m - 1 \neq 0$. Every factor in the product is therefore nonzero, and $\det(F) \neq 0$.

2. **Theorems used below (from class).**

(a) Isomorphisms and representing matrices. Let V be a finite-dimensional real vector space, let $T : V \to V$ be linear, and let $[T]_{\mathcal{B}}$ represent T in a basis $\mathcal{B}$. Then

$$T \text{ is an isomorphism} \iff [T]_{\mathcal{B}} \text{ is invertible.}$$

(b) Invertibility and the determinant. For every $A \in M_n(\mathbb{R})$,

$$A \text{ is invertible} \iff \det(A) \neq 0.$$

(This is the standard determinant criterion for invertibility.)

(c) Invariance of the determinant under change of basis. If P is invertible, then

$$\det(P^{-1}AP) = \det(A),$$

because

$$\det(P^{-1}AP) = \det(P^{-1}) \det(A) \det(P) = \det(A).$$

We can now solve the problem.

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EXERCISE 12.9 CONTINUED

Write $A = A(\alpha, \beta)$. Define

$$U = \Big\{ x \in \mathbb{R}^n : x_{p+1} = \cdots = x_n = 0, \quad \sum_{i=1}^{p} x_i = 0 \Big\},$$

$$W = \Big\{ x \in \mathbb{R}^n : x_1 = \cdots = x_p = 0, \quad \sum_{i=p+1}^{n} x_i = 0 \Big\}.$$

Clearly U, W are subspaces, since they are defined by homogeneous linear equations in the coordinates.

We first prove that $Ax = (1-\alpha)x$ for every $x \in U$. Let $x \in U$. Then $x_{p+1} = \cdots = x_n = 0$ and $\sum_{j=1}^{p} x_j = 0$.

If $1 \le i \le p$, the structure of row i of A (diagonal entry 1, entries α elsewhere in its block, and entries β between blocks) gives

$$(Ax)_i = 1 \cdot x_i + \sum_{\substack{1 \le j \le p \\ j \ne i}} \alpha x_j + \sum_{j=p+1}^{n} \beta x_j.$$

The last sum vanishes because $x_{p+1} = \cdots = x_n = 0$, so

$$(Ax)_i = x_i + \alpha \sum_{\substack{1 \le j \le p \\ j \ne i}} x_j.$$

Now

$$\sum_{\substack{1 \le j \le p \\ j \ne i}} x_j = \sum_{j=1}^{p} x_j - x_i = 0 - x_i = -x_i,$$

and hence

$$(Ax)_i = x_i + \alpha(-x_i) = (1-\alpha)x_i.$$

If $p+1 \le i \le n$, the coefficients of coordinates $1, \ldots, p$ in this row are all β (the second block of x is zero). Thus

$$(Ax)_i = \sum_{j=1}^{p} \beta x_j = \beta \sum_{j=1}^{p} x_j = 0 = (1-\alpha)x_i.$$

Therefore $Ax = (1-\alpha)x \in U$. In particular, U is A-invariant, and the restriction $A|_U$ is multiplication by $(1-\alpha)$. The same argument gives $Ay = (1-\alpha)y$ for every $y \in W$. Thus W is also A-invariant and $A|_W = (1-\alpha)\mathrm{Id}_W$.

Define

$$u = (\underbrace{1, \ldots, 1}_{p}, \underbrace{0, \ldots, 0}_{q}), \qquad v = (\underbrace{0, \ldots, 0}_{p}, \underbrace{1, \ldots, 1}_{q}), \qquad S = \mathrm{span}\{u, v\}.$$

We next compute Au and Av.

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EXERCISE 12.9 CONTINUED

For every $1 \le i \le p$,

$$(Au)_i = 1 \cdot 1 + \sum_{\substack{1 \le j \le p \\ j \ne i}} \alpha \cdot 1 + \sum_{j=p+1}^{n} \beta \cdot 0 = 1 + (p-1)\alpha.$$

and, for every $p+1 \le i \le n$,

$$(Au)_i = \sum_{j=1}^{p} \beta \cdot 1 + \Big(1 \cdot 0 + \sum_{\substack{p+1 \le j \le n \\ j \ne i}} \alpha \cdot 0\Big) = p\beta.$$

Therefore

$$Au = \big(1 + (p-1)\alpha\big)u + p\beta\, v.$$

Similarly,

$$Av = q\beta\, u + \big(1 + (q-1)\alpha\big)v.$$

Thus S is A-invariant, and its representing matrix in the basis (u, v) is

$$B = \begin{pmatrix} 1 + (p-1)\alpha & q\beta \\ p\beta & 1 + (q-1)\alpha \end{pmatrix}.$$

We claim that

$$\mathbb{R}^n = (U \oplus W) \oplus S.$$

First, $U \cap W = \{0\}$, since a vector in both subspaces must have every coordinate zero. Also, if $au + bv \in U \oplus W$, the sum of its first-block coordinates is $ap = 0$, so $a = 0$; the sum of its second-block coordinates is $bq = 0$, so $b = 0$. Hence $S \cap (U \oplus W) = \{0\}$.

Next, every $x \in \mathbb{R}^n$ admits a decomposition $x = x_{UW} + x_S$ with $x_{UW} \in U \oplus W$ and $x_S \in S$. Write

$$s_1 = \sum_{i=1}^{p} x_i, \qquad s_2 = \sum_{i=p+1}^{n} x_i,$$

and define

$$x_S = \frac{s_1}{p}u + \frac{s_2}{q}v \in S, \qquad x_{UW} = x - x_S.$$

The sum of the first-block coordinates of x_{UW} is $s_1 - s_1 = 0$, and the second-block sum is $s_2 - s_2 = 0$, so $x_{UW} \in U \oplus W$. This proves the direct-sum decomposition.

Choose a basis $\mathcal{B}$ by concatenating bases of U and W and the basis (u, v) of S. In this basis, the matrix of A has the block diagonal form

$$(1-\alpha)I_{n-2} \oplus B.$$

Let P be the change-of-coordinates matrix from $\mathcal{B}$ to the standard basis. Then

$$P^{-1}AP = (1-\alpha)I_{n-2} \oplus B.$$

Combining facts (b) and (c),

$$\det(A) = \det(P^{-1}AP) = \det\big((1-\alpha)I_{n-2} \oplus B\big).$$

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EXERCISE 12.9 CONTINUED

The determinant of a block diagonal matrix is the product of the determinants of its blocks (this follows by cofactor expansion, or by a suitable factorization and multiplicativity $\det(AB) = \det(A)\det(B)$). Therefore

$$\det(A) = \det\big((1-\alpha)I_{n-2}\big) \cdot \det(B) = (1-\alpha)^{n-2}\det(B).$$

Compute $\det(B)$:

$$\det(B) = \big(1+(p-1)\alpha\big)\big(1+(q-1)\alpha\big) - pq\,\beta^2.$$

and hence

$$\boxed{\det(A) = (1-\alpha)^{n-2}\Big(\big(1+(p-1)\alpha\big)\big(1+(q-1)\alpha\big) - pq\,\beta^2\Big)}.$$

By fact (a), T is an isomorphism if and only if its representing matrix A in the standard basis is invertible. By fact (b), this is equivalent to $\det(A) \neq 0$. Thus

$$T \text{ is an isomorphism} \iff \det(A) \neq 0 \iff (1-\alpha)^{n-2} \neq 0 \text{ and } \det(B) \neq 0.$$

Since $\alpha, \beta \in \mathbb{R}$ and $n \geq 3$, $(1-\alpha)^{n-2} \neq 0$ holds if and only if $\alpha \neq 1$. The exact condition is therefore

$$\boxed{\alpha \neq 1} \quad \text{and} \quad \boxed{\big(1+(p-1)\alpha\big)\big(1+(q-1)\alpha\big) \neq pq\,\beta^2}.$$

EXERCISE 12.10

For $s \in \mathbb{R}$, define $T_s : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T_s(v) = A_s v, \qquad A_s = \begin{pmatrix} 3 & 1 \\ s & 2 \end{pmatrix}.$$

The triangle Δ has vertices $(0,0)$, $(1,0)$, $(0,1)$. Please find the vertices and area of its image, please determine when T_s is invertible, and please describe the image when $s = 6$.

WORKED SOLUTION

The images of the vertices are $(0,0)$, $(3,s)$, $(1,2)$. The original triangle has area $1/2$, and the area scale factor is $|\det A_s| = |3\cdot 2 - 1\cdot s| = |6-s|$. Hence the image has area $|6-s|/2$. Also, T_s is invertible exactly when $6 - s \neq 0$, namely when $s \neq 6$. When $s = 6$, the columns satisfy $(3,6)^T = 3(1,2)^T$. Every image is therefore a multiple of $(1,2)^T$, and the triangle collapses to a line segment on $y = 2x$, with zero area. For $s < 6$, orientation is preserved; for $s > 6$, it is reversed. For example, at $s = 1$ the image has area $5/2$.

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EXERCISE 12.11 EXAM QUESTION

1. Let $A \in M_n(\mathbb{R})$ satisfy $\text{rank}(A^2) < \text{rank}(A)$. Please prove that there exists $v \in \mathbb{R}^n$ such that $A^2v = 0$ and $Av \neq 0$.

2. Let $A, B \in M_n(\mathbb{R})$. Please prove that

$$\text{rank}(A) + \text{rank}(B) \leq n + \text{rank}(AB).$$

Please deduce that if $\text{rank}(A) + \text{rank}(B) > n$, then $AB \neq 0$.

3. Let $N \in M_n(\mathbb{R})$ satisfy $N^2 = 0$ and $\text{rank}(N) = r$. Please find $\det(I_n + tN)$ and $\text{rank}(I_n + tN)$ for every $t \in \mathbb{R}$.

WORKED SOLUTION

1. If $Av = 0$, then $A^2v = A(Av) = 0$. Hence $\ker A \subseteq \ker A^2$. By the rank-nullity theorem and the given inequality,

$$\begin{aligned}\dim \ker A^2 &= n - \text{rank}(A^2) \\ &> n - \text{rank}(A) = \dim \ker A.\end{aligned}$$

Thus the inclusion is strict: $\ker A \subsetneq \ker A^2$. Choose $v \in \ker A^2 \setminus \ker A$. Membership in the first kernel gives $A^2v = 0$, while exclusion from the second gives $Av \neq 0$. This vector remains nonzero after one application of A and becomes zero after two applications.

2. Consider the restricted map

$$T : \text{Im}\, B \longrightarrow \mathbb{R}^n, \qquad T(u) = Au.$$

Its image is exactly $\text{Im}(AB)$: every $u \in \text{Im}\, B$ has the form $u = Bv$, so $T(u) = ABv$; conversely, every vector ABv arises in this way. Its kernel is $\text{Im}\, B \cap \ker A$. Applying rank-nullity to T gives

$$\text{rank}(B) = \dim(\text{Im}\, B \cap \ker A) + \text{rank}(AB).$$

However,

$$\dim(\text{Im}\, B \cap \ker A) \leq \dim \ker A = n - \text{rank}(A).$$

Substitute and rearrange:

$$\begin{aligned}\text{rank}(B) &\leq n - \text{rank}(A) + \text{rank}(AB), \\ \text{rank}(A) + \text{rank}(B) &\leq n + \text{rank}(AB).\end{aligned}$$

If the ranks sum to more than n, then $\text{rank}(AB) > 0$, so AB is not the zero matrix.

3. First compute the two products:

$$\begin{aligned}(I_n + tN)(I_n - tN) &= I_n - tN + tN - t^2N^2 = I_n, \\ (I_n - tN)(I_n + tN) &= I_n + tN - tN - t^2N^2 = I_n.\end{aligned}$$

Thus $I_n + tN$ is invertible for every $t \in \mathbb{R}$, with inverse $I_n - tN$. Immediately,

$$\text{rank}(I_n + tN) = n.$$

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EXERCISE 12.11 EXAM QUESTION CONTINUED

To compute the determinant, note that $N^2 = 0$ implies $\operatorname{Im} N \subseteq \ker N$. Hence $r \leq n - r$, or $2r \leq n$. Choose a basis $(u_1, \ldots, u_r)$ of $\operatorname{Im} N$ and extend it to a basis $(u_1, \ldots, u_r, w_1, \ldots, w_{n-2r})$ of $\ker N$. For each j, choose v_j with $Nv_j = u_j$.

The list

$$\mathcal{B} = (u_1, \ldots, u_r, w_1, \ldots, w_{n-2r}, v_1, \ldots, v_r)$$

is a basis. Indeed, applying N to a vanishing linear combination gives a combination of the u_j, so every coefficient of a v_j is zero. The remaining coefficients vanish because they multiply a basis of $\ker N$. Thus the list contains n linearly independent vectors. In this basis,

$$[N]_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & I_r \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad [I_n + tN]_{\mathcal{B}} = \begin{pmatrix} I_r & 0 & tI_r \\ 0 & I_{n-2r} & 0 \\ 0 & 0 & I_r \end{pmatrix}.$$

A block of size zero is simply omitted. This is upper triangular with every diagonal entry equal to 1. Since the determinant is preserved under change of basis,

$$\det(I_n + tN) = 1 \qquad \text{for every } t \in \mathbb{R}.$$

Neither answer depends on r: the determinant is 1 and the rank is n.

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